Lecture 9: Trees, Cutsets and Fundamental Loops

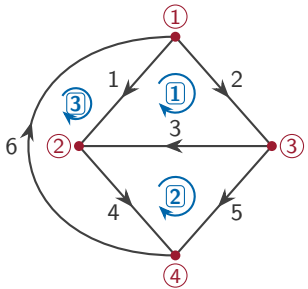
EE201 – Circuit Theory I

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current voltage, cutsets, loops tree nodes

Review: two matrices for one graph



$n = 4, \quad b = 6$

Incidence matrix (rows: nodes ①, ②, ③)

$$A = \begin{array}{c} \\ ① \\ ② \\ ③ \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & -1 \\ -1 & 0 & -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

Mesh matrix (rows: meshes 1, 2, 3)

$$M = \begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} -1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & -1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 \end{bmatrix} \end{array}$$

$n - 1 = 3$ independent KCL equations, $\quad b - n + 1 = 3$ independent KVL equations

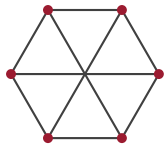
Every row of A is orthogonal to every row of M: $\quad AM^T = \mathbf{0}$

Review: node voltages and mesh currents

	node voltages	**mesh currents**
works for	any connected graph	planar, nonseparable
number of unknowns	$n-1$	$b-n+1$
holds automatically	KVL: $\mathbf{v} = A^T\mathbf{e}$	KCL: $\mathbf{i} = M^T\mathbf{i}_M$
left to write	KCL: $A\mathbf{i} = \mathbf{0}$	KVL: $M\mathbf{v} = \mathbf{0}$

Tellegen: $\mathbf{i}^T\mathbf{v} = 0$

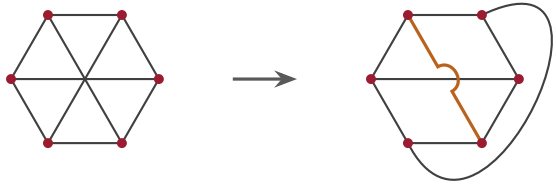
Nonplanar circuits



crossings at the centre

Can this graph be redrawn without crossings?

Nonplanar circuits

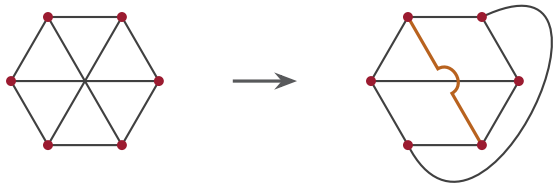


crossings at the centre

jumper wire: still one hop

No: at least one hop remains $\Rightarrow$ no meshes, no mesh matrix.

Nonplanar circuits



crossings at the centre

jumper wire: still one hop

No: at least one hop remains $\Rightarrow$ no meshes, no mesh matrix.

Question

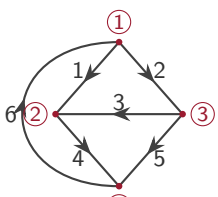
Without meshes, how do we find $n-1$ independent KCL and $b-n+1$ independent KVL equations?

Today: a method that needs no planar drawing and works for any connected graph.

Trees

Tree

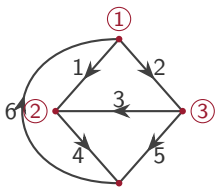
A subgraph T of a connected graph G is a tree if it contains all the nodes of G, is connected, and contains no loops.



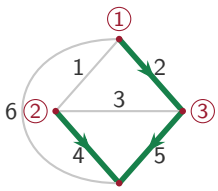
Trees

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graph G

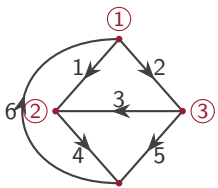


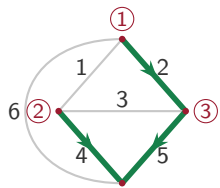
$\{2, 4, 5\}$: **tree?**

Trees

Tree

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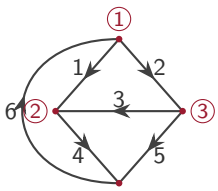




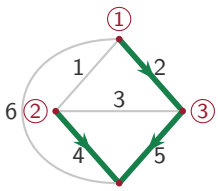
Trees

Tree

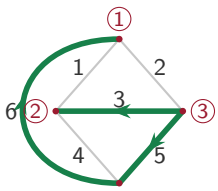
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graph G



$\{2, 4, 5\}$: **tree**

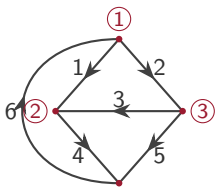


$\{3, 5, 6\}$: **tree?**

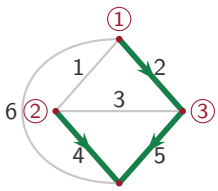
Trees

Tree

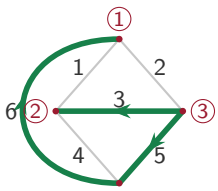
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graph G



$\{2, 4, 5\}$: **tree**

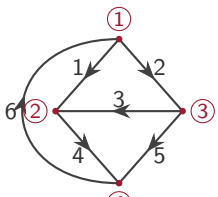


$\{3, 5, 6\}$: **tree**

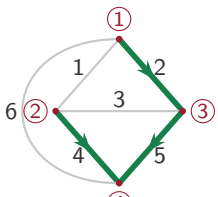
Trees

Tree

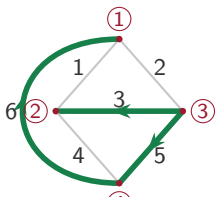
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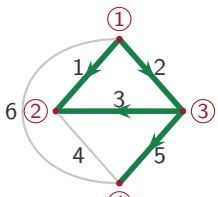
graph G



$\{2, 4, 5\}$: **tree**



$\{3, 5, 6\}$: **tree**

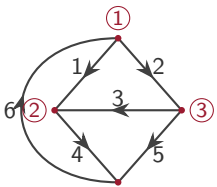


$\{1, 2, 3, 5\}$: **tree?**

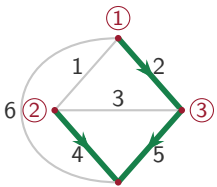
Trees

Tree

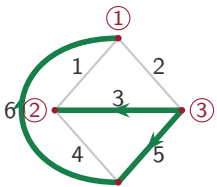
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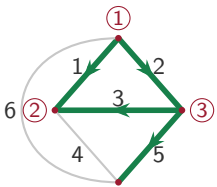
graph G



$\{2, 4, 5\}$: **tree**



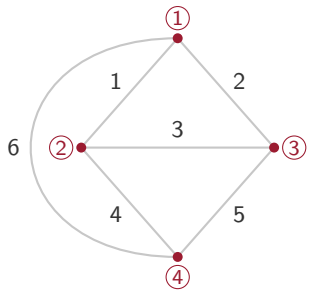
$\{3, 5, 6\}$: **tree**



$\{1, 2, 3, 5\}$: **not a tree**

Branches 1, 2, 3 form a loop. A graph has many trees.

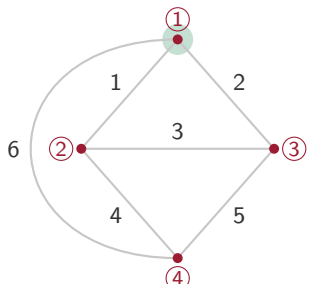
How many branches does a tree have?



$n = 4$ nodes, $b = 6$ branches

How many branches does a tree have?

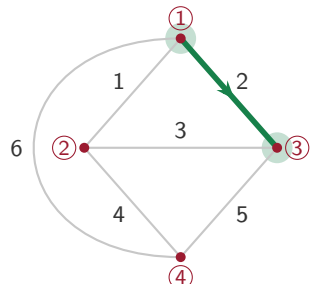
- Start from one node; add tree branches one at a time.



start at node ①

How many branches does a tree have?

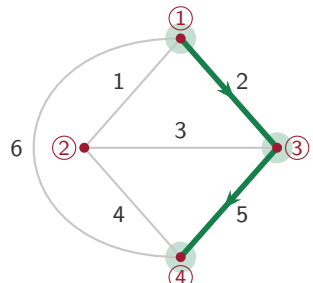
- Start from one node; add tree branches one at a time.
- Each new branch must reach a new node — otherwise it closes a loop.



branch 2 reaches node ③

How many branches does a tree have?

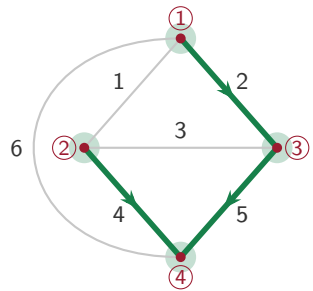
- Start from one node; add tree branches one at a time.
- Each new branch must reach a new node — otherwise it closes a loop.



branch 5 reaches node ④

How many branches does a tree have?

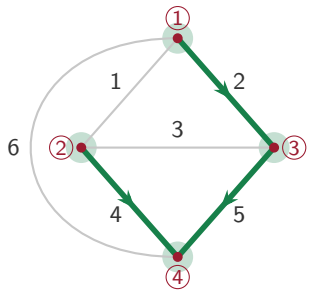
- Start from one node; add tree branches one at a time.
- Each new branch must reach a new node — otherwise it closes a loop.



branch 4 reaches node ②

How many branches does a tree have?

- Start from one node; add tree branches one at a time.
- Each new branch must reach a new node — otherwise it closes a loop.
- Each branch adds one node. All n nodes are reached after $n-1$ branches.

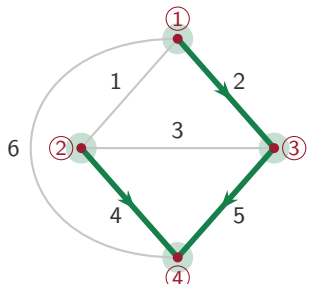


4 nodes, 3 tree branches: $n-1=3$

How many branches does a tree have?

- Start from one node; add tree branches one at a time.
- Each new branch must reach a new node — otherwise it closes a loop.
- Each branch adds one node. All n nodes are reached after $n-1$ branches.

The other $b-n+1$ branches are left out.



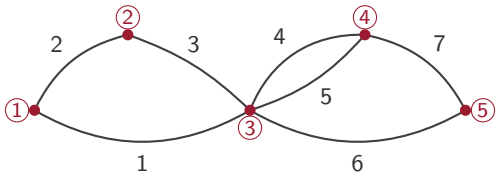
left out: 1, 3, 6 $(b-n+1=3)$

Cutsets

Cutset

A set C of branches of a connected graph is a cutset if

1. removing all of C leaves the graph unconnected, and
2. removing all of C except any one of them leaves it connected.

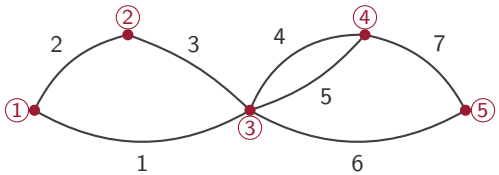


Cutsets

Cutset

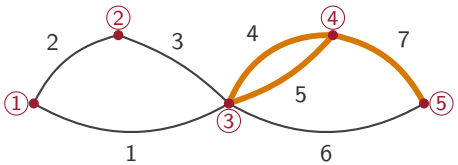
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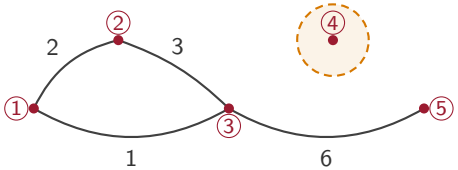
In words: removing the set cuts the graph into two pieces, and every branch in the set is needed for that.

Example: cutsets



$$C_1 = \{4, 5, 7\}$$

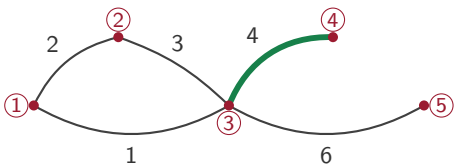
Example: cutsets



$C_1 = \{4, 5, 7\}$

(1) Remove all three: node ④ is cut off. ✓

Example: cutsets



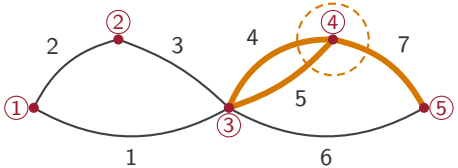
$C_1 = \{4, 5, 7\}$

(1) Remove all three: node ④ is cut off. ✓

(2) Put back one, e.g. 4: node ④ is reached again. Same for 5 or 7. ✓

$\Rightarrow$ C_1 is a cutset.

Example: cutsets



$C_1 = \{4, 5, 7\}$

(1) Remove all three: node ④ is cut off. ✓

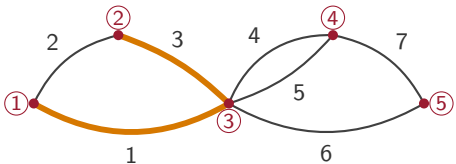
(2) Put back one, e.g. 4: node ④ is reached again. Same for 5 or 7. ✓

$\Rightarrow$ C_1 is a cutset.

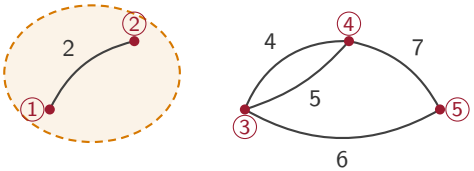
All branches back: the closed surface around node ④ crosses exactly 4, 5, 7.

Example: cutsets

$$C_2 = \{1, 3\}$$



Example: cutsets



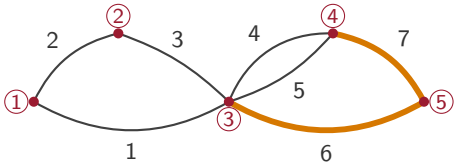
$C_2 = \{1, 3\}$

(1) Remove both: nodes ①, ② are cut off. ✓

(2) Put back 1 or 3: connected again. ✓

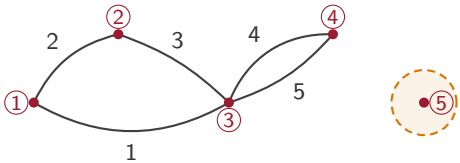
$\Rightarrow$ C_2 is a cutset.

Example: cutsets



$$C_3 = \{6, 7\}$$

Example: cutsets



$C_3 = \{6, 7\}$

(1) Remove both: node ⑤ is cut off. ✓

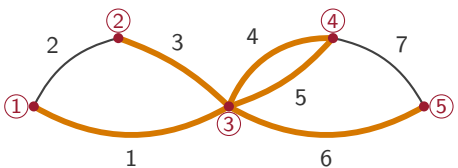
(2) Put back 6 or 7: connected again. ✓

$\Rightarrow$ C_3 is a cutset.

Cutsets can have different sizes: C_1 has 3 branches, C_2 and C_3 have 2.

Example: not every set is a cutset

$C = \{1, 3, 4, 5, 6\}$: a cutset?



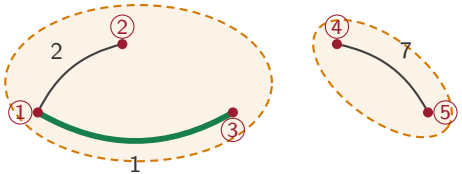
Example: not every set is a cutset



$C = \{1, 3, 4, 5, 6\}$: a cutset?

(1) Remove all five: three pieces. ✓

Example: not every set is a cutset



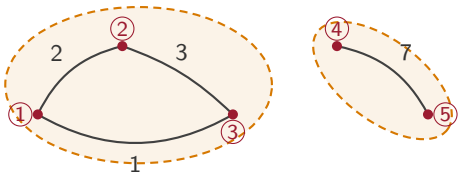
$C = \{1, 3, 4, 5, 6\}$: a cutset?

(1) Remove all five: three pieces. ✓

(2) Put back 1: still unconnected. ×

$\Rightarrow$ not a cutset: some branches are not needed.

Example: not every set is a cutset



$C = \{1, 3, 4, 5, 6\}$: a cutset?

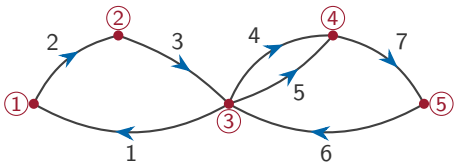
(1) Remove all five: three pieces. ✓

(2) Put back 1: still unconnected. ×

$\Rightarrow$ not a cutset: some branches are not needed.

Branches 1, 3 are extra: $\{4, 5, 6\}$ alone is a cutset.

KCL on a cutset



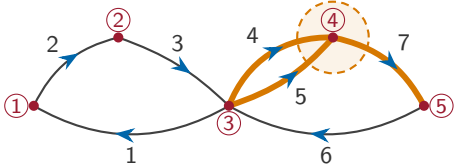
branch	1	2	3	4	5	6	7
current (A)	2	2	2	1	2	3	3

Branch currents in the arrow directions.

They satisfy KCL at every node, e.g. node ③:

$$i_1 + i_4 + i_5 - i_3 - i_6 = 2+1+2-2-3 = 0.$$

KCL on a cutset



branch	1	2	3	4	5	6	7
current (A)	2	2	2	1	2	3	3

$C_1 = \{4, 5, 7\}$, surface around node ④:

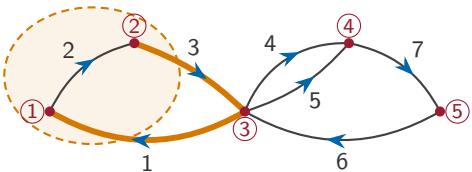
leaving: i_7; entering: i_4, i_5

$$i_7 - i_4 - i_5 = 3 - 1 - 2 = 0 \checkmark$$

Here this is just KCL at node ④.

KCL on a cutset: the cutset currents add up to zero
(+ leaving the surface, − entering).

KCL on a cutset



branch	1	2	3	4	5	6	7
current (A)	2	2	2	1	2	3	3

$C_2 = \{1, 3\}$, surface around nodes ①, ②:

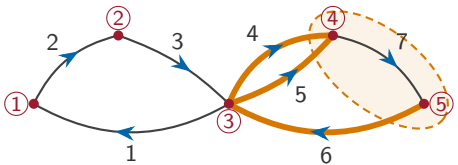
leaving: i_3; entering: i_1

$$i_3 - i_1 = 2 - 2 = 0 \checkmark$$

KCL at ①: $i_2 - i_1 = 0$; at ②: $i_3 - i_2 = 0$. Add them: i_2 is inside the surface and cancels.

KCL on a cutset: the cutset currents add up to zero
(+ leaving the surface, − entering).

KCL on a cutset



branch	1	2	3	4	5	6	7
current (A)	2	2	2	1	2	3	3

$\{4, 5, 6\}$, surface around nodes ④, ⑤:

leaving: i_6; entering: i_4, i_5

$$i_6 - i_4 - i_5 = 3 - 1 - 2 = 0 \checkmark$$

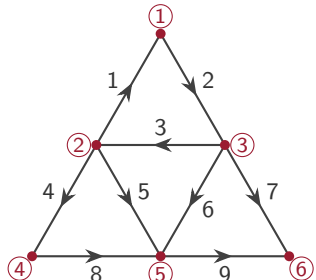
Sum of KCL at ④ and ⑤: i_7 cancels.

KCL on a cutset: the cutset currents add up to zero
(+ leaving the surface, − entering).

Fundamental cutsets

Fundamental cutset

Take a tree T and a tree branch β. Exactly one cutset contains β and no other tree branch: the fundamental cutset of β. Its direction is that of β.



graph G: $n = 6$, $b = 9$

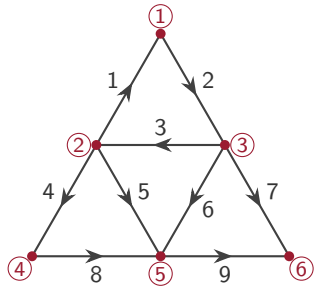
Fundamental cutsets

Fundamental cutset

Take a tree T and a tree branch β. Exactly one cutset contains β and no other tree branch: the fundamental cutset of β. Its direction is that of β.

Cotree

The branches that are not in the tree.



graph G: $n = 6$, $b = 9$

Fundamental cutsets

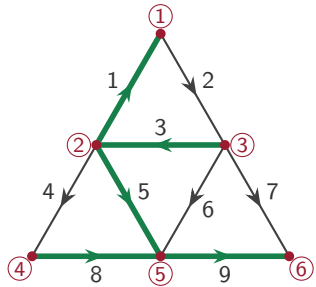
Fundamental cutset

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Cotree

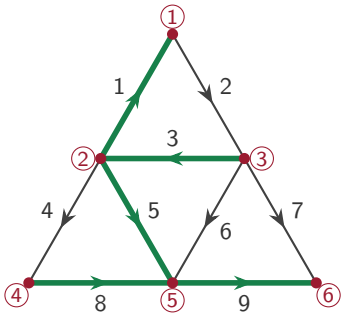
The branches that are not in the tree.

Tree $\{1, 3, 5, 8, 9\}$: $n - 1 = 5$. Cotree $\{2, 4, 6, 7\}$: $b - n + 1 = 4$.



graph G: $n = 6$, $b = 9$

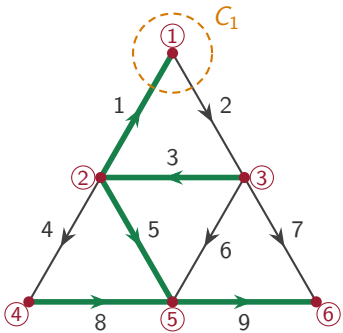
Example: fundamental cutsets of G



Remove one tree branch: the tree falls in two parts. The fundamental cutset joins them.

tree br.	cutset	KCL

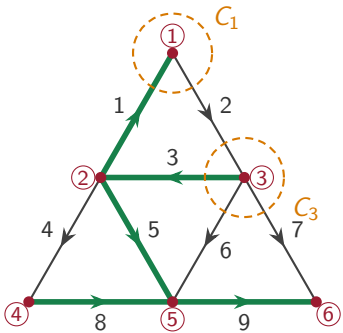
Example: fundamental cutsets of G



Remove one tree branch: the tree falls in two parts. The fundamental cutset joins them.

tree br.	cutset	KCL
1	$\{1, 2\}$	$i_1 - i_2 = 0$

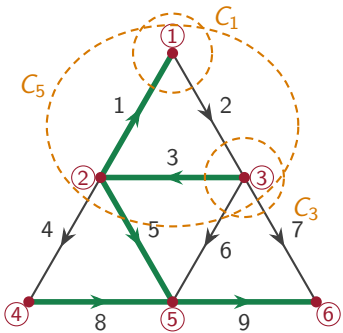
Example: fundamental cutsets of G



Remove one tree branch: the tree falls in two parts. The fundamental cutset joins them.

tree br.	cutset	KCL
1	$\{1, 2\}$	$i_1 - i_2 = 0$
3	$\{3, 2, 6, 7\}$	$i_3 - i_2 + i_6 + i_7 = 0$

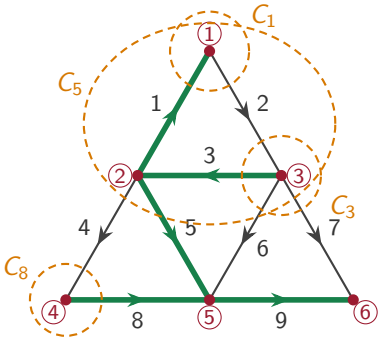
Example: fundamental cutsets of G



Remove one tree branch: the tree falls in two parts. The fundamental cutset joins them.

tree br.	cutset	KCL
1	$\{1,2\}$	$i_1 - i_2 = 0$
3	$\{3,2,6,7\}$	$i_3 - i_2 + i_6 + i_7 = 0$
5	$\{5,4,6,7\}$	$i_5 + i_4 + i_6 + i_7 = 0$

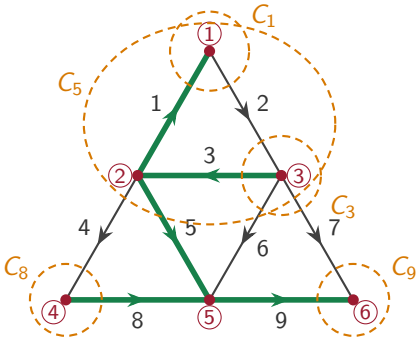
Example: fundamental cutsets of G



Remove one tree branch: the tree falls in two parts. The fundamental cutset joins them.

tree br.	cutset	KCL
1	$\{1,2\}$	$i_1 - i_2 = 0$
3	$\{3,2,6,7\}$	$i_3 - i_2 + i_6 + i_7 = 0$
5	$\{5,4,6,7\}$	$i_5 + i_4 + i_6 + i_7 = 0$
8	$\{8,4\}$	$i_8 - i_4 = 0$

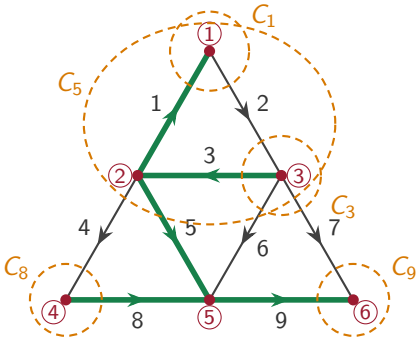
Example: fundamental cutsets of G



Remove one tree branch: the tree falls in two parts. The fundamental cutset joins them.

tree br.	cutset	KCL
1	$\{1,2\}$	$i_1 - i_2 = 0$
3	$\{3,2,6,7\}$	$i_3 - i_2 + i_6 + i_7 = 0$
5	$\{5,4,6,7\}$	$i_5 + i_4 + i_6 + i_7 = 0$
8	$\{8,4\}$	$i_8 - i_4 = 0$
9	$\{9,7\}$	$i_9 + i_7 = 0$

Example: fundamental cutsets of G



Remove one tree branch: the tree falls in two parts. The fundamental cutset joins them.

tree br.	cutset	KCL
1	$\{1,2\}$	$i_1 - i_2 = 0$
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5	$\{5,4,6,7\}$	$i_5 + i_4 + i_6 + i_7 = 0$
8	$\{8,4\}$	$i_8 - i_4 = 0$
9	$\{9,7\}$	$i_9 + i_7 = 0$

$n - 1 = 5$ KCL equations, independent: each has its own tree branch current.

Cotree currents fix the tree currents

Each fundamental cutset has one tree branch. Solve for it:

$$i_1 = i_2, \qquad i_3 = i_2 - i_6 - i_7, \qquad i_5 = -i_4 - i_6 - i_7, \qquad i_8 = i_4, \qquad i_9 = -i_7$$

Know the $b - n + 1$ cotree currents $\Rightarrow$ know all branch currents.

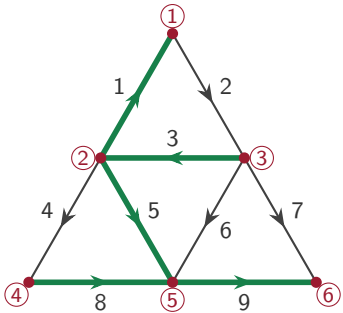
Fundamental loops

Fundamental loop

Each cotree branch closes exactly one loop with the tree: that cotree branch plus the tree path between its ends. It is the fundamental loop of that cotree branch. Its direction is that of the cotree branch.

KVL around it: $+v_k$ if the arrow of branch k points along the walk, $-v_k$ if against.

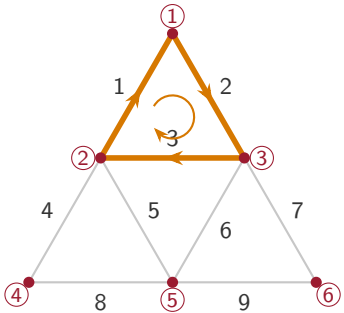
Example: fundamental loops of G



tree $\{1, 3, 5, 8, 9\}$, cotree $\{2, 4, 6, 7\}$

cotree br.	loop	KVL

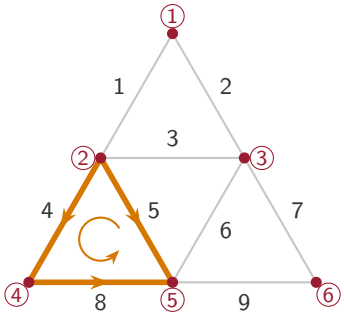
Example: fundamental loops of G



tree $\{1, 3, 5, 8, 9\}$, cotree $\{2, 4, 6, 7\}$

cotree br.	loop	KVL
2	$\{2, 3, 1\}$	$v_2 + v_3 + v_1 = 0$

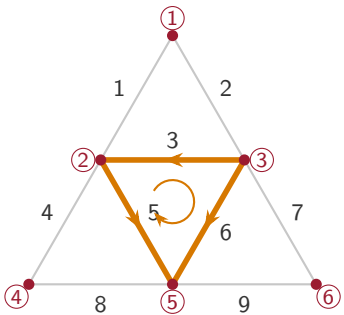
Example: fundamental loops of G



tree $\{1, 3, 5, 8, 9\}$, cotree $\{2, 4, 6, 7\}$

cotree br.	loop	KVL
2	$\{2, 3, 1\}$	$v_2 + v_3 + v_1 = 0$
4	$\{4, 8, 5\}$	$v_4 + v_8 - v_5 = 0$

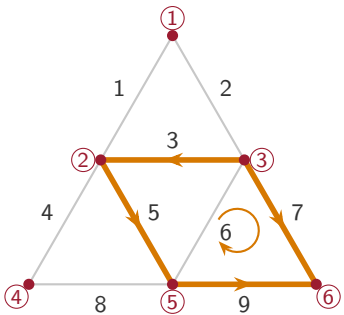
Example: fundamental loops of G



tree $\{1, 3, 5, 8, 9\}$, cotree $\{2, 4, 6, 7\}$

cotree br.	loop	KVL
2	$\{2, 3, 1\}$	$v_2 + v_3 + v_1 = 0$
4	$\{4, 8, 5\}$	$v_4 + v_8 - v_5 = 0$
6	$\{6, 5, 3\}$	$v_6 - v_5 - v_3 = 0$

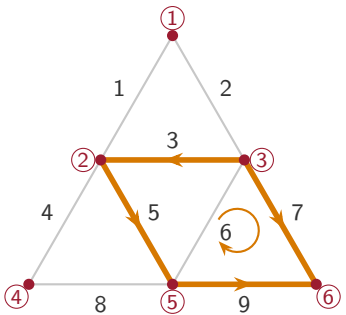
Example: fundamental loops of G



tree $\{1, 3, 5, 8, 9\}$, cotree $\{2, 4, 6, 7\}$

cotree br.	loop	KVL
2	$\{2, 3, 1\}$	$v_2 + v_3 + v_1 = 0$
4	$\{4, 8, 5\}$	$v_4 + v_8 - v_5 = 0$
6	$\{6, 5, 3\}$	$v_6 - v_5 - v_3 = 0$
7	$\{7, 9, 5, 3\}$	$v_7 - v_9 - v_5 - v_3 = 0$

Example: fundamental loops of G



tree $\{1, 3, 5, 8, 9\}$, cotree $\{2, 4, 6, 7\}$

cotree br.	loop	KVL
2	$\{2, 3, 1\}$	$v_2 + v_3 + v_1 = 0$
4	$\{4, 8, 5\}$	$v_4 + v_8 - v_5 = 0$
6	$\{6, 5, 3\}$	$v_6 - v_5 - v_3 = 0$
7	$\{7, 9, 5, 3\}$	$v_7 - v_9 - v_5 - v_3 = 0$

$b - n + 1 = 4$ KVL equations, independent: each has its own cotree branch voltage.

Tree voltages fix the cotree voltages

Each fundamental loop has one cotree branch. Solve for it:

$$v_2 = -v_1 - v_3, \qquad v_4 = v_5 - v_8, \qquad v_6 = v_3 + v_5, \qquad v_7 = v_3 + v_5 + v_9$$

Know the $n - 1$ tree voltages $\Rightarrow$ know all branch voltages.

Any connected graph

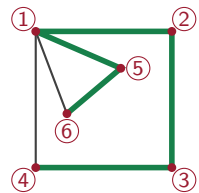
	how many	gives
tree branches	$n-1$	fundamental cutsets: the KCL equations
cotree branches	$b-n+1$	fundamental loops: the KVL equations

Any connected graph

	how many	gives
tree branches	$n-1$	fundamental cutsets: the KCL equations
cotree branches	$b-n+1$	fundamental loops: the KVL equations

No drawing needed: works for nonplanar graphs.

It also fixes the separable case met earlier: with this tree, the two fundamental loops are the square and the triangle — each a single loop.



Looking ahead

	unknowns	fix every
node voltages	$n-1$	branch voltage
tree branch voltages	$n-1$	branch voltage
mesh currents	$b-n+1$	branch current
cotree branch currents	$b-n+1$	branch current

Trees give the general version — for any connected graph.

Summary

- **Tree**: all nodes, connected, no loop; $n-1$ branches. The rest: the **cotree**, $b-n+1$ branches.
- **Cutset**: minimal set of branches that cuts the graph in two. KCL holds on it.
- **Fundamental cutsets** (one per tree branch): $n-1$ independent KCL equations. Cotree currents fix the tree currents.
- **Fundamental loops** (one per cotree branch): $b-n+1$ independent KVL equations. Tree voltages fix the cotree voltages.
- Works for any connected graph — planar or not, separable or not.

Next lecture

Duality

The dual of a planar graph, and the dual of a circuit.

Questions?